

Waiz Khan

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EDUCATION

Johns Hopkins University

M.S.E. in Data Science

- Coursework: Machine Translation, Artificial Intelligence, Deep Learning, Applied Statistics and Data Analysis

Baltimore, MD

Aug. 2024 – May 2027

GD Goenka University

B.Tech. in Computer Science & Engineering

Gurugram, India

Aug. 2019 – Nov. 2023

RESEARCH EXPERIENCE

Research Assistant

Dec. 2025 – Present

Johns Hopkins University, Advisor: Prof. Philipp Koehn

Baltimore, MD

- Design and validate evaluation metrics for LLM representations: an intrinsic multilinguality metric tested across 17 open models on NTREX-128 and Belebele, against 30+ preregistered predictions and held out model families
- Falsified the metric under optimization pressure: training against it directly produced a collapse shortcut that scored well without exhibiting the property being measured, isolating a construct validity failure rather than an implementation bug
- Run the experimental program on a Slurm GPU cluster, from representation extraction at scale to intervention studies across 6 training objectives and 3 seeds; author the reports behind an NSF proposal in development with Prof. Koehn
- Earlier phase: benchmarked 15+ decoder architectures and learning rules for hieroglyph to text translation on a 61k example corpus, reaching BLEU-4 30.3

Research Assistant

Aug. 2022 – Nov. 2022

GD Goenka University, Advisor: Dr. Yogesh Kumar

Gurugram, India

- Studied chronic school absenteeism on the IHDS-II national survey of 40k+ students; logistic regression, KNN and random forest models reached 74% accuracy

EXPERIENCE

Founding Engineer & Co-Founder

Apr. 2026 – Present

Arzaic LLC, Iris: an LLM agent platform for heart failure care

Baltimore, MD

Built through the Johns Hopkins Pava Center accelerator

- Architected a staged LLM agent pipeline (gate → route → retrieve → generate → guard) with an embedding based semantic router, agentic retrieval over a versioned multimodal patient record, and multimodal RAG that captions clinical artifacts before embedding them
- A safety judge fails closed on every response and safety evaluation runs as a merge gate, so no change ships without passing it; a nightly agent drafts care plan revisions for clinician review

PROJECTS

AttendOpt | *Python, scikit-learn, Optuna, DoWhy*

2026

- Rebuilt a 2022 absenteeism study over the full 47,027 student IHDS-II cohort: gradient boosting reaches ROC-AUC 0.841 against a 0.699 logistic baseline, DoWhy confirms the distance barrier causally (ATE 0.14 per SD, refutations pass), and an integer program doubles expected impact per fixed budget against random targeting

Glyphos | *Python, PyTorch, scikit-learn*

Aug. 2026

- Evaluation infrastructure for decipherment research over eleven ancient language corpora; an audit of the standard test splits found roughly 34% near duplicate leakage, the class of error that inflates every downstream result, motivating 26 frozen partitions and a CI gate blocking pretrained weights

ExoSAGE | *Python, FastAPI, D3*

Jul. 2026

- Point in time screening of NASA TOI transit candidates with false discovery rate control and walk forward backtesting, plus mixed integer telescope hour allocation under correlated risk; 11 test suites over checksummed snapshots and real TESS light curves

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, C++, Swift, Bash

LLMs & Agents: agent pipelines and tool use, RAG and embedding retrieval, prompt engineering, agent evaluation harnesses and safety gating, Hugging Face Transformers

ML: PyTorch, scikit-learn, XGBoost, CatBoost; transformers, CNNs; causal inference (DoWhy); experimental design, preregistered evaluation

Systems & Data: Node.js, Next.js, SQLite/PostgreSQL, pandas/NumPy/SciPy, Slurm HPC clusters, Git, Docker